

Shridhan Suman

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Professional Summary

Results-driven AI Engineer with over a year of experience building scalable LLM applications and autonomous AI agents. Skilled in Python, LangGraph, FastAPI, and vector databases, with a strong background in data engineering and Machine Learning. Proven problem-solving capabilities demonstrated through 350+ LeetCode problems.

Education

VIT, Chennai <i>B.Tech in Computer Science and Engineering - CGPA: 8.43/10.0</i>	Expected June 2027
Bal Vidya Mandir Sr. Sec. School <i>Class XII (PCM) - 91.2%</i>	2022 Sambhal

Experience

Applied AI Intern - CSRBOX	Dec 2025 - Jan 2026
- Completed an intensive Applied AI program through IBM SkillsBuild, gaining hands-on expertise in NLP, Prompt Engineering, Agentic AI architectures, and end-to-end machine learning workflows. - Designed and delivered a real-world AI solution by performing data preprocessing, model development, evaluation, and iterative optimization based on mentor feedback, earning final project approval and an Applied AI certification.	

Technical Skills

AI & ML: Agentic AI, RAG, LangChain, LangGraph, CrewAI, Hugging Face, LLM Evals, NLP, Data Classification, Prompt Engineering

Languages & Frameworks: Python, JavaScript, MERN Stack, FastAPI, Streamlit

Data & Vector Stores: Pinecone, Supabase, Vector Embeddings, FAISS, IBM Cognos, Data Pipelines (Cleaning, Preprocessing, Imputation)

Tools & APIs: Groq, Azure Speech SDK (TTS), Serper API, GitHub, Google Colab, Clerk, Git

Projects

StudyMate: AI-Powered Adaptive Learning System	github.com/StudyMate
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Tech Stack: Llama 3, RAG, Pinecone, Supabase, FastAPI, Azure TTS, LangChain, React

- Engineered a robust RAG pipeline using **Llama 3** and **Pinecone** to enable a highly contextual AI Tutor with specialized Socratic and Feynman modes, significantly reducing hallucinations.
- Developed an AI Studio pipeline utilizing **Azure TTS** to dynamically generate personalized revision podcasts by aggregating user quiz mistakes and conceptual gaps.
- Optimized LLM inference to achieve **low latency** and improved cost efficiency by reducing API token consumption through strict context window management and advanced prompt engineering.
- Built an adaptive testing engine that autonomously generates dynamic quizzes from uploaded materials, evaluating user responses and providing real-time feedback.
- Designed a persistent performance tracking ecosystem featuring a React-based analytics dashboard, GitHub-style consistency heatmaps, and automated AI-driven learning paths.

LangGraph Multi-Agent Chatbot	github.com/Chatbot
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Tech Stack: LangGraph, Groq, Serper API, SQLite, Streamlit, Python, Pydantic

- Built a multi-agent chatbot using **LangGraph**, orchestrating specialized agents through state graphs and conditional routing for task execution.
- Integrated **Serper API** and custom Python tools to enable real-time web search and external information retrieval during conversations.
- Used **Groq** for low-latency LLM inference and implemented conversation memory with **SQLite** to maintain context across multiple interactions.
- Developed an interactive **Streamlit** interface for chat, tool execution, and agent workflow visualization, improving usability and debugging.

Task-Specific AI Agents

Tech Stack: LangGraph, Python, Serper API, Structured Outputs, Prompt Engineering

- Engineered automated, multi-step task workflows using **LangGraph**, orchestrating specialized worker agents for targeted task execution (e.g., Researcher, Code Writer, Content Reviewer).
- Implemented **structured outputs** and strict schema definitions (Pydantic) to automate complex task execution, guaranteeing the syntactical validity of generated code and consistent formatting.
- Grounded agent reasoning in real-time web data via the **Serper API**, significantly reducing hallucinations and enhancing factual accuracy during research tasks.
- Designed complex system prompts and autonomous tool-calling loops, allowing agents to iteratively plan, execute, and self-correct their task outputs without human intervention.